

MEDIASTINUM

High-yield anatomy notes with larger source-PDF diagrams and image explanations

Definition and position

The mediastinum is the broad median partition of the thoracic cavity between the right and left pleural cavities. It contains the thoracic viscera and structures except the lungs.

- It is covered on both sides by mediastinal pleura.
- It occupies the central compartment of the thoracic cavity.
- In the living body it is mobile because many structures are hollow and joined by loose connective tissue.

Boundaries

Anterior: sternum.

Posterior: bodies of thoracic vertebrae and intervertebral discs.

Superior: superior thoracic aperture.

Inferior: diaphragm.

Lateral: right and left mediastinal pleura.

Major contents of the mediastinum

- Thymus.
- Heart enclosed in the pericardial sac.
- Large arteries and veins such as aorta, pulmonary trunk and superior vena cava.
- Trachea and principal bronchi.
- Esophagus and thoracic duct.
- Nerves: phrenic nerves, vagus nerves, sympathetic trunks and cardiac nerves.
- Mediastinal lymph nodes.

Study order

Definition and boundaries.

General contents.

Divisions and transverse thoracic plane.

Contents of each subdivision.

Applied anatomy and radiology.

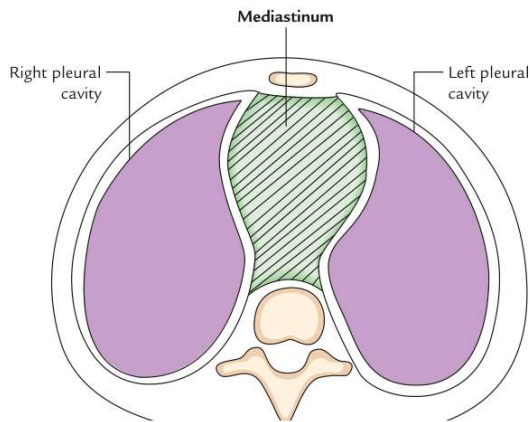
Exam scoring points

Always mention that lungs are excluded.

Use T4/sternal angle as the key plane.

Write contents in groups: vessels, nerves, tubes, lymphatics.

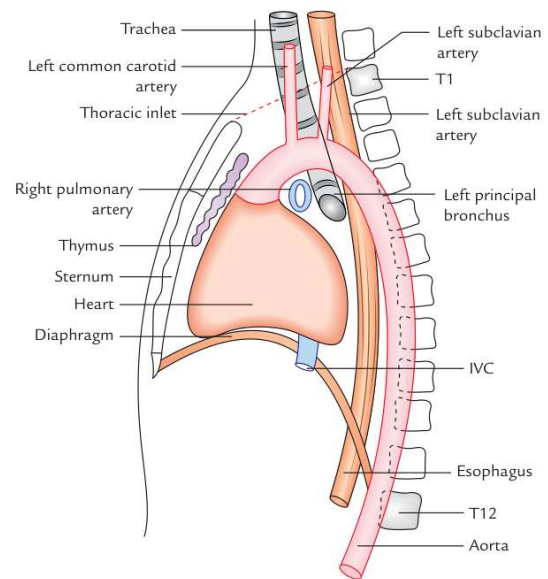
Add one clinical correlation in long answers.

Fig. 19.1 - Position in thoracic cross-section

The green central area is the mediastinum. It separates the two pleural cavities and therefore acts as the anatomical partition between the right and left lungs.

Fig. 19.2 - Boundaries and contents

1. Thymus.
2. Heart enclosed in the pericardial sac.



This sagittal diagram shows the mediastinum from the thoracic inlet to the diaphragm. It includes structures such as thymus, heart, trachea, esophagus, aorta and major vessels.

MEDIASTINAL SURFACES AND LATERAL RELATIONS

Before reading the lateral diagram

Identify the lung root first.

Phrenic nerve is anterior to the root.

Vagus nerve is posterior to the root.

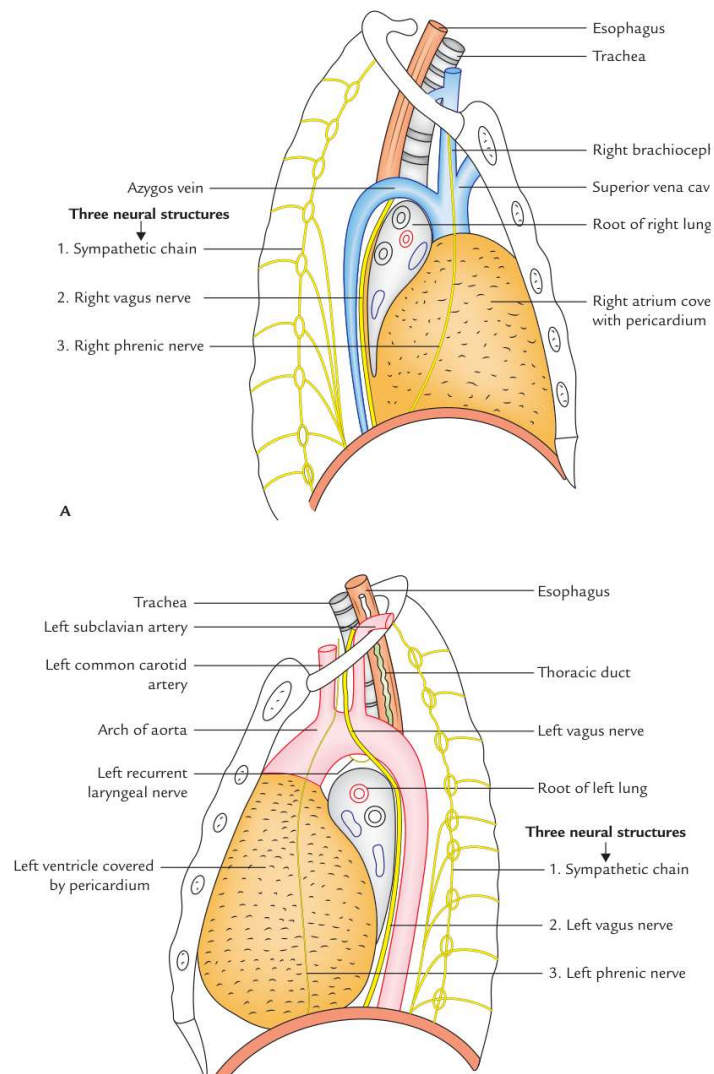
Sympathetic chain is still more posterior.

Side dominance

Right side: SVC, right atrium and azygos arch.

Left side: arch of aorta, thoracic duct and left ventricle.

Left recurrent laryngeal nerve hooks near arch of aorta.

Fig. 19.3 - Mediastinal structures seen from right and left sides

The figure compares the right and left lateral relations. Remember the nerve rule: phrenic nerve passes anterior to the lung root, vagus nerve passes posterior to it, and sympathetic chain lies still more posteriorly. The right side is venous; the left side is aortic.

Right side of mediastinum

Commonly related to trachea, esophagus, right brachiocephalic vein, SVC, root of right lung, right atrium covered by pericardium and arch of azygos vein.

- Right phrenic nerve: anterior to root of lung.
- Right vagus nerve: posterior to root of lung.
- Right sympathetic chain: posterior vertical neural structure.

Left side of mediastinum

Commonly related to trachea, esophagus, left subclavian artery, left common carotid artery, arch of aorta, thoracic duct, root of left lung and left ventricle covered by pericardium.

- Left recurrent laryngeal nerve hooks under the arch of aorta.
- Left phrenic nerve passes anterior to the root of lung.
- Left vagus nerve passes posterior to the root of lung.

DIVISIONS OF THE MEDIASTINUM

Transverse thoracic plane

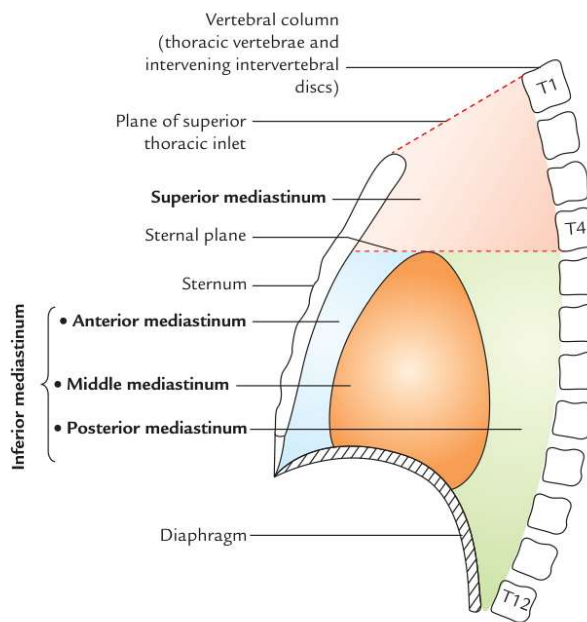
For description, the mediastinum is divided by an imaginary transverse thoracic plane. This plane passes through the sternal angle anteriorly and the lower border of T4 vertebra or T4-T5 disc posteriorly.

- Above the plane: superior mediastinum.
- Below the plane: inferior mediastinum.
- The inferior mediastinum is divided by the pericardium into anterior, middle and posterior mediastina.

How to visualize the compartments

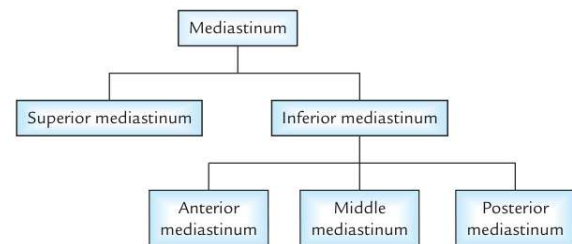
Imagine the sternal plane as a horizontal shelf. Everything above it is superior mediastinum. Below it, the pericardial sac acts like a central box: the space in front is anterior, the box itself is middle, and the space behind is posterior.

Fig. 19.4 - Divisions in sagittal view



The sternal plane separates superior and inferior mediastinum. Below this, the pericardium divides the inferior mediastinum into anterior, middle and posterior parts.

Flowchart 19.1 - Subdivisions



A quick exam layout: mediastinum -> superior and inferior; inferior -> anterior, middle and posterior mediastinum.

Reading Fig. 19.4 quickly

The main landmark is the transverse thoracic plane. In a sagittal diagram, it runs from the sternal angle to T4. The pericardium then divides the inferior mediastinum into three practical spaces used in radiology, anatomy and surgery.

Visual memory

Superior mediastinum lies above the sternal angle plane.
Anterior mediastinum is the narrow prepericardial space.
Middle mediastinum is the pericardial compartment.
Posterior mediastinum lies behind the pericardium.

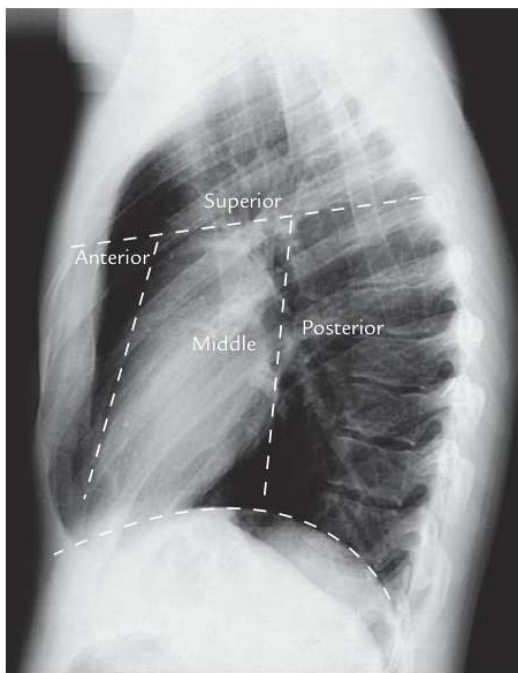
Radiology memory

Retrosternal clear space: anterior mediastinum.
Cardiac shadow: middle mediastinum.
Retrocadiac area: posterior mediastinum.
Above the sternal plane: superior mediastinum.

Compartment	Best visual cue
Superior	Above sternal angle/T4 plane
Anterior	In front of pericardium
Middle	Heart and pericardium
Posterior	Behind pericardium, in front of vertebral bodies

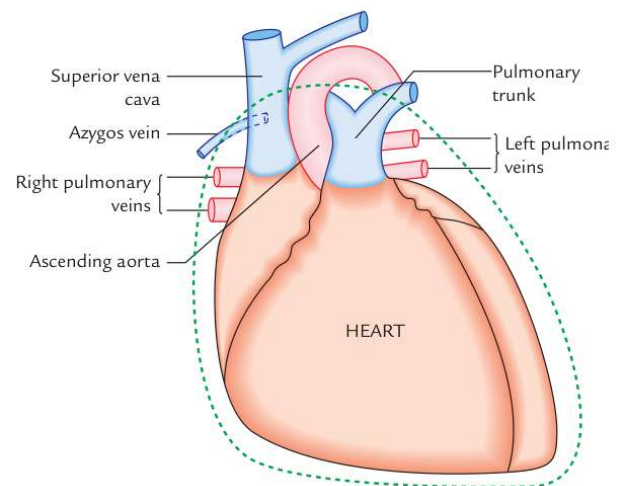
Fig. 19.5 - Lateral chest X-ray subdivision

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On a lateral chest radiograph, the region above the sternal plane represents superior mediastinum. Retrosternal, cardiac and retrocardiac regions correspond broadly to anterior, middle and posterior mediastina.

Fig. 19.8 - Middle mediastinum projection



The middle mediastinum contains the pericardium, heart and roots of great vessels. The dotted outline shows the pericardial area and therefore the practical boundary of the middle mediastinum.

Subdivision	Position / boundary idea	Main memory point
Superior mediastinum	Above transverse thoracic plane	Great vessels, trachea, esophagus, thymus and important nerves.

Anterior mediastinum	In front of pericardium	Smallest and narrowest part; contains loose tissue, lymph nodes and sternopericardial ligaments.
Middle mediastinum	Pericardium and its contents	Largest component is the heart; roots of great vessels lie here.
Posterior mediastinum	Behind pericardium	Main passage region for esophagus, aorta, thoracic duct, azygos system and autonomic nerves.

SUPERIOR MEDIASTINUM

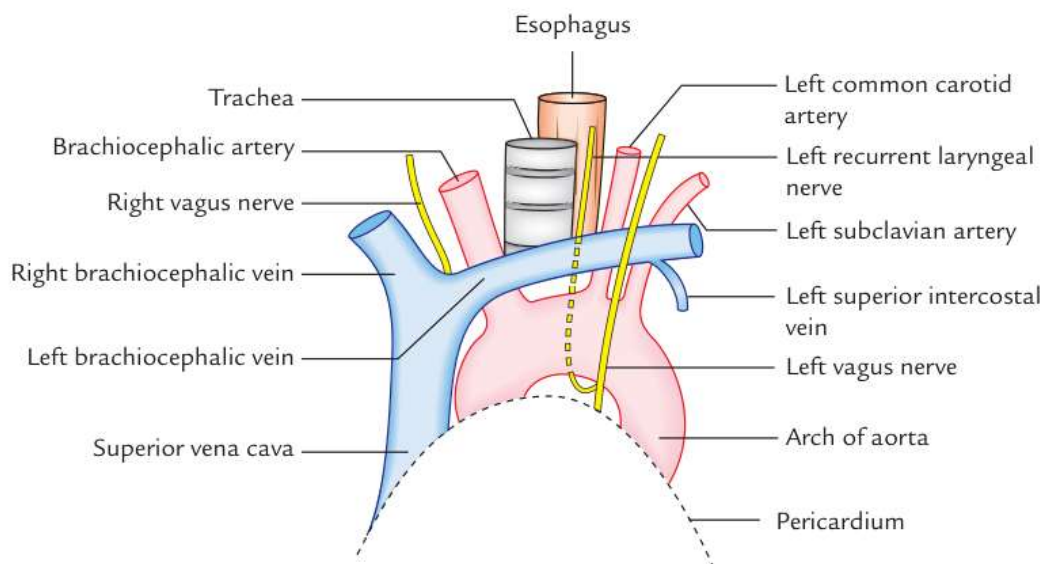
Boundaries

Anterior: manubrium sterni.
 Posterior: bodies of upper four thoracic vertebrae.
 Superior: plane of superior thoracic aperture.
 Inferior: transverse thoracic plane at sternal angle and lower border of T4.
 Lateral: mediastinal pleura.

Contents - grouped

Arteries: arch of aorta and its three branches.
 Veins: brachiocephalic veins, upper half of SVC and left superior intercostal vein.
 Nerves: phrenic, vagus, sympathetic trunks, cardiac nerves and left recurrent laryngeal nerve.
 Others: thymus, lymph nodes, thoracic duct, trachea, esophagus, sternohyoid, sternothyroid and longus colli.

Fig. 19.6 - Superior mediastinum in dissection

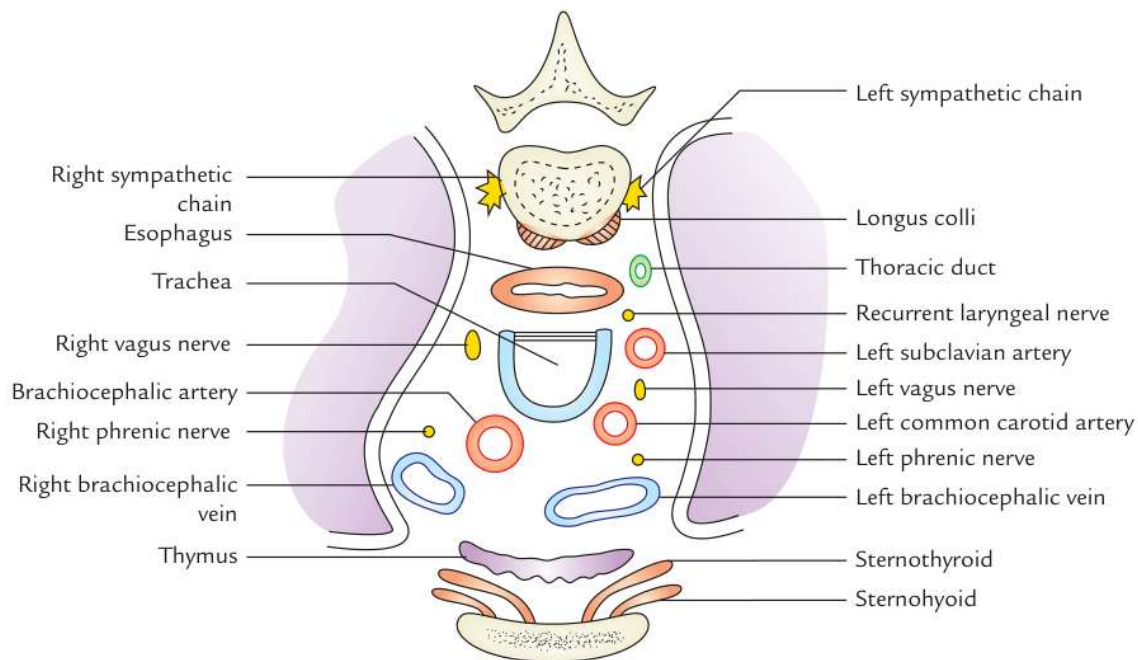


This dissection view is especially important because it shows the rule that great veins lie anterior to the great arteries. The trachea and esophagus are posterior, and the left recurrent laryngeal nerve lies near the arch of aorta.

Surgical orientation in superior mediastinum

- From anterior to posterior: thymus -> large veins -> large arteries -> trachea -> esophagus and thoracic duct -> sympathetic trunks.
- Potential dead space: veins are mostly on the right side while large arteries are mostly on the left. In increased blood flow, veins can expand whereas large arteries do not expand much; masses may project into this available space.

Fig. 19.7 - Transverse section of superior mediastinum



In transverse section, structures can be arranged from anterior to posterior: sternohyoid/sternothyroid and thymus, brachiocephalic veins, arteries, trachea, esophagus with thoracic duct, and sympathetic chains. This is a useful surgical orientation pattern.

ANTERIOR MEDIASTINUM

Boundaries

Anterior: body of sternum.
 Posterior: pericardium enclosing heart.
 Superior: transverse thoracic plane.
 Inferior: diaphragm.
 Lateral: mediastinal pleura.

Contents

Loose areolar tissue.
 Superior and inferior sternopericardial ligaments.
 Three or four lymph nodes.
 Mediastinal branches of internal thoracic arteries.
 Lower part of thymus in children.

Clinical note - anterior mediastinum

The anterior mediastinum is a very narrow space. It is continuous above with the superior mediastinum and with the pretracheal space of the neck. Therefore, infection from the pretracheal region of the neck may spread down into the anterior mediastinum. In children, the

lower part of the thymus may occupy this space.

Point	Reason it matters in exams
Smallest mediastinal subdivision	Anterior mediastinum is often described as a narrow space in front of the pericardium.
Sternopericardial ligaments	These fibrous bands connect pericardium to sternum and help anchor the pericardial sac.
Child vs adult	A lower portion of thymus may be present in children but usually regresses with age.

MIDDLE MEDIASTINUM

Boundaries

Anterior: anterior mediastinum.
 Posterior: posterior mediastinum.
 Superior: superior mediastinum.
 Inferior: diaphragm.
 Lateral: mediastinal pleura on each side.

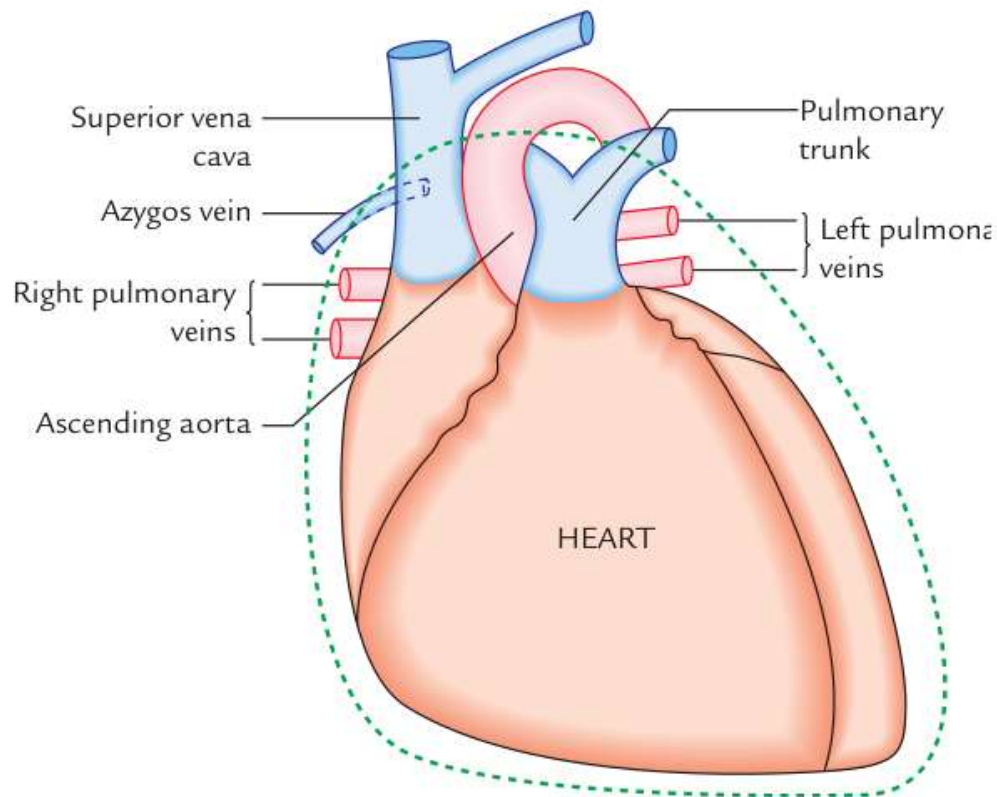
Contents

Heart and pericardium.
 Arteries: ascending aorta, pulmonary trunk and pericardiophrenic arteries.
 Veins: lower half of SVC, terminal azygos vein and pulmonary veins.
 Nerves: phrenic nerves and deep cardiac plexus.
 Lymph nodes: tracheobronchial nodes.
 Tubes: bifurcation of trachea and right and left principal bronchi.

Why the middle mediastinum is central

The middle mediastinum is the functional core of the thoracic cavity because it encloses the heart and pericardium. It is also the area where the great vessels enter and leave the heart, so vascular enlargement, pericardial pathology and cardiac enlargement commonly alter the mediastinal outline on imaging.

Middle mediastinum group	Examples to write in exams
Pericardial structures	Heart and pericardium
Arteries	Ascending aorta, pulmonary trunk, pericardiophrenic arteries
Veins	Lower half of SVC, terminal azygos vein, pulmonary veins
Nerves and plexus	Phrenic nerves and deep cardiac plexus
Airway and nodes	Tracheal bifurcation, principal bronchi, tracheobronchial nodes

Fig. 19.8 - Main contents of middle mediastinum

The middle mediastinum is essentially the pericardial compartment. The heart is the largest structure of the mediastinum. The superior vena cava, ascending aorta, pulmonary trunk, pulmonary veins, terminal azygos vein, pericardium and tracheal bifurcation are clinically important landmarks here.

Clinical relevance of middle mediastinum

Since the heart and pericardium occupy this region, enlargement of the heart, pericardial effusion, or dilation of great vessels can produce widening of the mediastinum on radiographs. The phrenic nerves pass with pericardiophrenic vessels on the fibrous pericardium, so irritation or injury may affect diaphragmatic movement.

POSTERIOR MEDIASTINUM

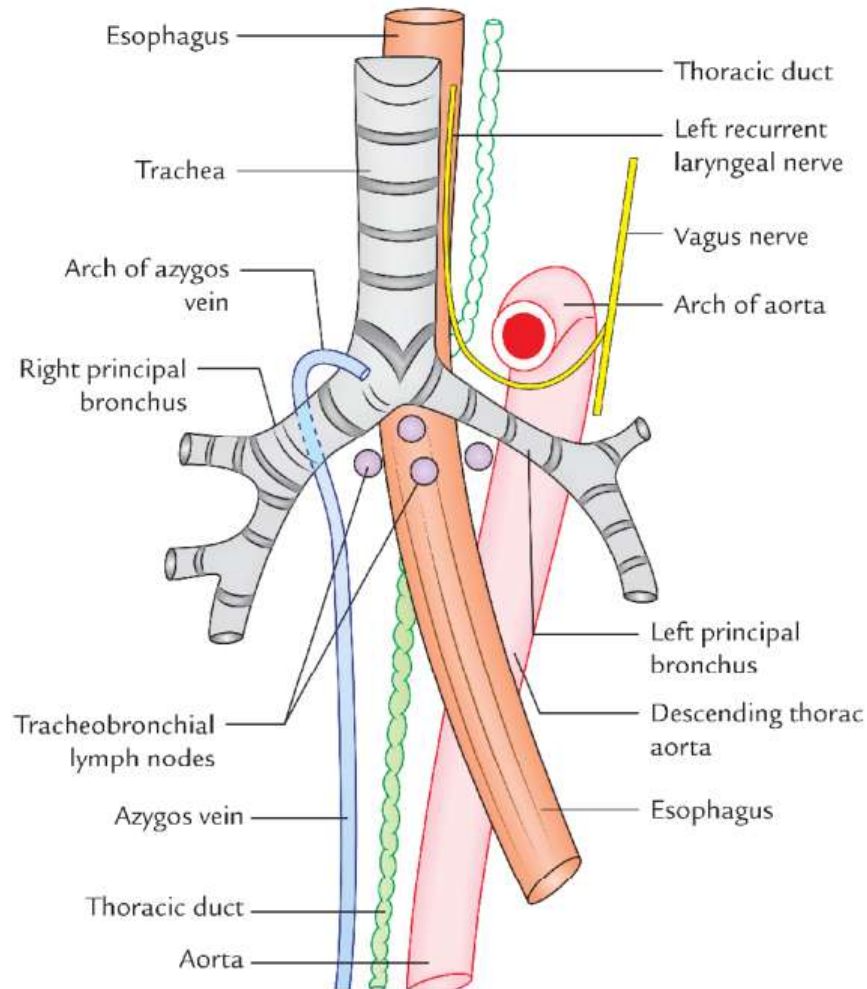
Boundaries

Anterior: pericardium and its contents, bifurcation of trachea and pulmonary vessels.
 Posterior: bodies of lower eight thoracic vertebrae and intervertebral discs.
 Superior: transverse thoracic plane.
 Inferior: diaphragm.
 Lateral: mediastinal pleura.

Contents

Esophagus.
 Thoracic duct.
 Descending thoracic aorta and its branches.
 Azygos, hemiazygos and accessory hemiazygos veins.
 Vagus nerves.
 Sympathetic trunks and splanchnic nerves.

Posterior mediastinal lymph nodes.

Fig. 19.9 - Posterior mediastinum and related structures

This figure is a compact map of posterior mediastinum. Note the esophagus, thoracic duct, descending thoracic aorta, azygos vein, vagus nerve and left recurrent laryngeal nerve. A key landmark is that the arch of aorta passes over the left bronchus, whereas the azygos vein arches over the right bronchus.

Clinical spread through posterior mediastinum

The posterior mediastinum communicates with spaces in the neck above and with fascial planes below. Deep infections of the neck may descend into the posterior mediastinum. Pus from the posterior mediastinum may also enter the psoas sheath through a funnel-shaped opening and track down to the thigh near the femoral triangle.

APPLIED ANATOMY OF MEDIASTINUM

Condition / sign

Anatomical basis

Mediastinitis	Inflammation of loose connective tissue of mediastinum; deep neck infections can spread into thorax through fascial spaces.
Subcutaneous emphysema at root of neck	Air may escape into loose mediastinal connective tissue after esophageal perforation and then track upward to the neck.
Mediastinal syndrome	Compression by tumor, cyst or enlarged structures may cause venous obstruction, dyspnea, dysphagia, hoarseness or vertebral erosion.
Widened mediastinum	May result from hemorrhage after trauma, malignant lymph node enlargement or cardiac enlargement.
Mediastinal shift	Shift to affected side in collapse/atelectasis; shift to healthy side in pneumothorax or hydrothorax.

Symptoms from compression

Venous engorgement: compression/obstruction of SVC.

Dyspnea: compression of trachea.

Dysphagia: compression of esophagus.

Hoarseness of voice: compression of left recurrent laryngeal nerve.

Back pain or deformity: erosion of vertebral bodies.

Radiology and bedside clues

Lateral chest X-ray helps identify superior, anterior, middle and posterior compartments.

Mediastinal widening suggests vascular injury, lymph node mass, tumor or cardiac enlargement.

Tracheal position in the suprasternal notch helps detect mediastinal shift.

A right-sided pressure lesion may exploit the superior mediastinal dead space.

Golden facts to remember

- Mediastinum contains all thoracic viscera and structures except the lungs.
- Largest structure in mediastinum: heart.
- Commonest cause of mediastinal widening: cardiac enlargement/hypertrophy.
- Mediastinum is a dynamic, pliable and movable septum.
- Anterior mediastinum is the narrowest mediastinal subdivision.
- Posterior mediastinum is the key passage region for esophagus, aorta, thoracic duct and azygos system.

QUICK EXAM REVISION

Question prompt	Answer pattern
Define mediastinum.	Median thoracic partition between the pleural cavities containing all thoracic viscera except lungs.
Plane dividing superior and inferior mediastinum?	Transverse thoracic plane through sternal angle and lower border of T4/T4-T5 disc.
Anterior-to-posterior order in superior mediastinum?	Thymus, large veins, large arteries, trachea, esophagus with thoracic duct, sympathetic trunks.
Most important content of middle mediastinum?	Pericardium with heart and roots of great vessels.
Posterior mediastinum contents?	Esophagus, thoracic duct, descending aorta, azygos system, vagus nerves, sympathetic trunks/splanchnic nerves and nodes.
How to detect mediastinal shift clinically?	By palpating the trachea in the suprasternal notch.

Last-minute diagram strategy

For exams, draw a sagittal thorax with the sternal plane at the sternal angle and T4. Mark the region above as superior mediastinum. Draw the pericardium below the plane and label the space in front as anterior, the pericardial compartment as middle, and the region behind as posterior. Add one contents mnemonic beside each subdivision to make the diagram self-explanatory.

One-line mnemonics

Superior: great vessels + airway + food pipe + nerves.

Anterior: small space in front of pericardium.

Middle: heart, pericardium and roots of great vessels.

Posterior: passageway behind the heart.

Common diagram labels

Sternal angle / T4 plane.

Pericardium.

Trachea and esophagus.

Aorta and SVC.

Phrenic nerve anterior to root of lung.

Vagus nerve posterior to root of lung.

Final revision paragraph

The mediastinum should be understood as a three-dimensional central corridor rather than only a list of contents. The superior part connects the neck to the thorax, the middle part is built around the pericardium and heart, and the posterior part forms a vertical passage for the esophagus, thoracic duct, descending aorta and autonomic nerves. This is why infections, tumors, vascular injury and pressure effects can produce symptoms in the neck, thorax and even lower limb fascial pathways.